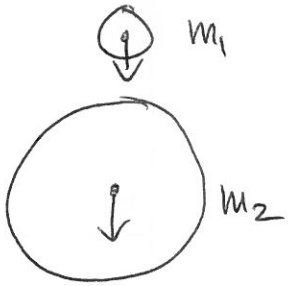
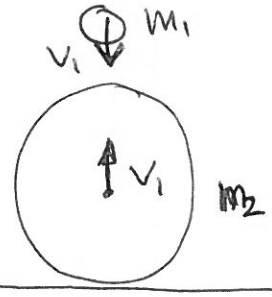
P5



Basketball collides with the ground and bounces back



$$m_1 g h_0 = \frac{1}{2} m_1 v_1^2$$

$$v_1^2 = 2 g h_0$$

Its velocity v_1 has the same magnitude as that of tennis ball right above it (as long as we can neglect ball's finite dimensions).

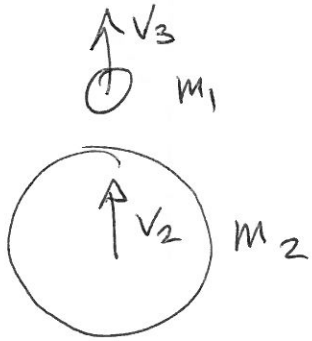
Thus $v_1 = \sqrt{2 g h_0}$

Momentum before collision of m_1 and m_2:

$$-m_1 v_1 + m_2 v_1 = P_{before}$$

$$E_{before} = \frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_1^2$$

After collision:



$$P_{after} = m_1 v_3 + m_2 v_2$$

$$E_{after} = \frac{1}{2} m_1 v_3^2 + \frac{1}{2} m_2 v_2^2$$

$$\Rightarrow \quad V_1(m_2 - m_1) = m_1 V_3 + m_2 V_2 \qquad (1) \text{ momentum}$$

$$\frac{1}{2}(m_1+m_2)V_1^2 = \frac{1}{2} m_1 V_3^2 + \frac{1}{2} m_2 V_2^2 \qquad (2) \text{ energy (elastic collisions!)}$$

From (1): $$V_2 = \frac{V_1(m_2-m_1) - m_1 V_3}{m_2}$$

From (2): $$(m_1+m_2)V_1^2 = m_1 V_3^2 + m_2 \left(\frac{V_1(m_2-m_1) - m_1 V_3}{m_2}\right)^2 =$$

$$= m_1 V_3^2 + \frac{(m_2-m_1)^2 V_1^2 + m_1^2 V_3^2 - 2m_1(m_2-m_1)V_1V_3}{m_2}$$

Multiply both sides by m_2 and re-arrange:

$$\underbrace{\left[m_2(m_1+m_2) - (m_2-m_1)^2\right]}_{3m_2m_1 - m_1^2} V_1^2 = m_1 m_2 V_3^2 + m_1^2 V_3^2 - 2m_1(m_2-m_1) \cdot V_1 V_3$$

Now use that $m_2 \gg m_1 \Rightarrow \frac{m_1}{m_2} \ll 1$ and can be neglected:

$$m_2 m_1 \left(3 - \cancel{\frac{m_1}{m_2}}\right) V_1^2 = m_1 m_2 \left\{ V_3^2 + \cancel{\frac{m_1}{m_2}} V_3^2 - 2V_1V_3 + 2\cancel{\frac{m_1}{m_2}} V_1 V_3 \right\}$$

Then $$3V_1^2 = V_3^2 - 2V_1V_3$$

$\Rightarrow$ quadratic equation for V_3: $V_3^2 - 2V_1V_3 - 3V_1^2 = 0$

$$V_3 = V_1 \pm \sqrt{V_1^2 + 3V_1^2}$$

Take positive root because V_3 is upward: $V_3 = 3V_1$!

Tennis ball rises up to h_1: $m_1 g h_1 = \frac{1}{2} m_1 V_3^2 = \frac{1}{2} m_1 g V_1^2$

$$\Rightarrow h_1 = 9h_0 = 13.5 \text{ m}$$